Supplementary Material to "Monopoly Insurance with Endogenous Information"

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This supplementary material to our paper "Monopoly Insurance with Endogenous Information" explores an alternative setup in which the cost of effort is monetary. More precisely, the expected utility of an agent with perceived risk $\tilde{\beta}$ who exerts effort e and buys an insurance contract (p, R) is given by

$$\tilde{\beta}u(w - p - D + R - e) + (1 - \tilde{\beta})u(w - p - e).$$

That is, here the cost of effort enters in the argument of the utility function as a monetary cost, instead of being additively separable as in the paper. The purpose of this supplementary material is to argue that the results in the paper are not driven by the additive structure we assume there. In particular, we will show that with CARA preferences and given certain technical assumptions (to be specified in condition S1 below),[1] a result that is the equivalent of Proposition 1 in the paper holds. We will also outline proofs for why also the other propositions and lemmas stated in the paper hold in the alternative setting studied here.

The ex ante utility of an agent exerting effort e and buying insurance contract (p_1, R_1) when receiving signal σ^h and buying insurance contract (p_2, R_2) when receiving signal σ^l is

$$\begin{aligned} U(e) &= \alpha\left[\beta^h(e)u(w - p_1 - D + R_1 - e) + (1 - \beta^h(e))u(w - p_1 - e)\right] \\ &\quad +(1-\alpha)\left[\beta^l(e)u(w - p_2 - D + R_2 - e) + (1 - \beta^l(e))u(w - p_2 - e)\right] \\ &= \alpha\left[\beta^h(e)\underline{u}_1(e) + (1 - \beta^h(e))\bar{u}_1(e)\right] + (1-\alpha)\left[\beta^l(e)\underline{u}_2(e) + (1 - \beta^l(e))\bar{u}_2(e)\right] \end{aligned} \tag{1}$$

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[1]The technical assumptions are satisfied, for example, if we assume either one of the signaling technologies described in Appendix A of the paper.

where we define $\bar{u}_i$ and $\underline{u}_i$ analogously to the paper. However, these variables are now functions of e and are no longer fixed by the offered contract. The rest of the setup, e.g. timing of events and assumptions on u, remains unchanged.

We will solve this new model by backwards induction.

1. Contract choice

The contract choice problem of the agent does not really change, relative to the analysis in the paper, because the effort choice was already determined previously. Hence, $\bar{u}_i(e)$ and $\underline{u}_i(e)$ are constant with respect to e at this point in time. Denoting by $\tilde{\beta}$ the agents perceived probability of suffering the loss D, we get the following.

The agent will remain uninsured if

$$\tilde{\beta} \leq \min\left(\frac{\bar{u}_0(e) - \bar{u}_2(e)}{\Delta_0(e) - \Delta_2(e)}, \frac{\bar{u}_0(e) - \bar{u}_1(e)}{\Delta_0(e) - \Delta_1(e)}\right).$$

The agent buys contract 2 if

$$\frac{\bar{u}_0(e) - \bar{u}_2(e)}{\Delta_0(e) - \Delta_2(e)} \leq \tilde{\beta} \leq \frac{\bar{u}_2(e) - \bar{u}_1(e)}{\Delta_2(e) - \Delta_1(e)}.$$

In the remaining cases, that is, when $\tilde{\beta}$ is high enough, the agent will buy the high coverage contract 1.

The notation is as in the paper: $\Delta_i(e) = \bar{u}_i(e) - \underline{u}_i(e)$ and a subscript 0 denotes the outside option of remaining uninsured (formally, $p_0 = R_0 = 0$). Furthermore, as in the paper, we let contract 1 be the contract with more coverage, i.e. $p_2 \leq p_1$ and $R_2 \leq R_1$. This implies $\Delta_1(e) \leq \Delta_2(e)$ for any $e \geq 0$.

2. Effort choice

We write utility as function of effort as

$$\begin{aligned} U(e) = \alpha \left[\beta_h(e)\underline{u}(\sigma^h, e) + (1 - \beta_h(e))\bar{u}(\sigma^h, e)\right] \\ + (1 - \alpha)\left[\beta_l(e)\underline{u}(\sigma^l, e) + (1 - \beta_l(e))\bar{u}(\sigma^l, e)\right], \end{aligned} \quad (2)$$

where $\underline{u}(\sigma^i, e)$ is the utility an agent gets in the case of an accident when buying the contract an agent with signal σ^i will optimally buy after exerting effort e. Similarly, $\bar{u}(\sigma^i, e)$ is the utility an agent will get in the no accident case when he bought the contract an agent should optimally buy after exerting effort e and receiving signal σ^i.

If an agent buys contract 1 after receiving a high signal and contract 2 after receiving a low signal, his first order condition in the effort choice problem is

$$\alpha {\beta_h}'(e)\left(\Delta_2(e)-\Delta_1(e)\right)+\alpha\left(\beta_h(e)\underline{u}_1'(e)+(1-\beta_h(e))\bar{u}_1'(e)\right)$$
$$+(1-\alpha)\left(\beta_l(e)\underline{u}_2'(e)+(1-\beta_l(e))\bar{u}_2'(e)\right)\leq 0 \quad \text{with “ = ” if } e>0. \quad (3)$$

Note that $\underline{u}_i'(e) < 0$ and $\bar{u}_i'(e) < 0$ as higher effort costs reduce disposable income. Consequently, the first term is positive while the second and third terms are negative. The second order condition of this maximization problem is

$$\alpha {\beta_h}''(e)\left(\Delta_2(e)-\Delta_1(e)\right)+2\alpha {\beta_h}'(e)\left(\Delta_2'(e)-\Delta_1'(e)\right)$$
$$+\alpha\left(\beta_h(e)\underline{u}_1''(e)+(1-\beta_h(e))\bar{u}_1''(e)\right)+(1-\alpha)\left(\beta_l(e)\underline{u}_2''(e)+(1-\beta_l(e))\bar{u}_2''(e)\right)\leq 0. \quad (4)$$

Because of the concavity of u, the second line is negative. The first term is negative by ${\beta_h}'' \leq 0$ that we used in the paper. But the second term is positive by the strict concavity of u. This implies that—in contrast to the setup in the paper—there could be multiple local maxima with $e > 0$. Whether this complication occurs depends on the properties of the signal technology and the utility function.

The possible existence of multiple interior local maxima complicates the analysis considerably. Some of the proofs in the paper still go through but for some we need more structure. The goal of this supplementary material is to illustrate that similar effects and results as the one described in the paper also exist if the effort cost is monetary. Therefore, we do not attempt a full-fledged analysis but restrict ourselves to a more structured environment in which all our results will go through. Nevertheless, we will point out in the remainder where additional complications would arise if we analyzed a more general framework. The more structured environment is given by the following condition.

Condition S1. *The utility function u has constant absolute risk aversion, i.e. $u(x) = -\mathbf{e}^{-\eta x}$ for some $\eta > 0$ (where $\mathbf{e}$ is the Euler number 2.718...). The signal technology is such that ${\beta_h}'(e) + \eta\beta_h(e)$ is strictly quasiconvex.*

The following lemma states that condition S1 solves the problem of multiple local interior maxima. It also shows that reasonable signal technologies satisfy the requirement in condition S1.

Lemma S1. *Under condition S1, there exists at most one $e > 0$ that satisfies (3) and (4). For both example technologies used in the paper, ${\beta_h}'(e) + \eta\beta_h(e)$ is strictly quasiconvex.*

Proof. With CARA preferences the first order condition (3) can be written as (after multiplying through with $\mathbf{e}^{-\eta(-w+e)}$)

$$\alpha\beta_h{}'(e)\left(-\mathbf{e}^{-\eta(-p_2)}+\mathbf{e}^{-\eta(-p_2-D+R_2)}+\mathbf{e}^{-\eta(-p_1)}-\mathbf{e}^{-\eta(-p_1-D+R_1)}\right)-\eta\alpha\beta_h(e)\mathbf{e}^{-\eta(-p_1-D+R_1)}$$
$$-\eta\alpha(1-\beta_h(e))\mathbf{e}^{-\eta(-p_1)}-\eta(1-\alpha)\beta_l(e)\mathbf{e}^{-\eta(-p_2-D+R_2)}-\eta(1-\alpha)(1-\beta_l(e))\mathbf{e}^{-\eta(-p_2)}=0.$$

Using $\beta=\alpha\beta_h(e)+(1-\alpha)\beta_l(e)$ and $1-\beta=\alpha(1-\beta_h(e))+(1-\alpha)(1-\beta_l(e))$, this can be rewritten as

$$\alpha\left(\beta_h{}'(e)+\eta\beta_h(e)\right)\left(-\mathbf{e}^{-\eta(-p_2)}+\mathbf{e}^{-\eta(-p_2-D+R_2)}+\mathbf{e}^{-\eta(-p_1)}-\mathbf{e}^{-\eta(-p_1-D+R_1)}\right)$$
$$=\eta\alpha\left(\mathbf{e}^{-\eta(-p_1)}-\mathbf{e}^{-\eta(-p_2)}\right)+\eta\beta\mathbf{e}^{-\eta(-p_2-D+R_2)}+\eta(1-\beta)\mathbf{e}^{-\eta(-p_2)}. \quad (5)$$

The only term including effort in the last equation is $(\beta_h{}'(e)+\eta\beta_h(e))$. All other terms are constant in the effort choice stage. By assumption, $(\beta_h{}'(e)+\eta\beta_h(e))$ is strictly quasiconvex. Hence, the first order condition is satisfied with equality at at most two positive effort levels. Since $U(e)$ is differentiable at every $e>0$, this implies that there can be at most one maximum with $e>0$.[2]

For the linear signal technology used in the numerical example of the paper, we have $\beta_h{}'(e)+\eta\beta_h(e)=e(\theta^h-\theta^l)(1-\alpha)\eta+(1-\alpha(1-\eta))\theta^h-(1-\alpha)(1-\eta)\theta^l$, which is linear and therefore strictly quasiconvex in e.

For the signal technology given as an example in the model section of the paper, we have

$$\eta\beta_h{}'(e)+\beta_h{}''(e)=\frac{(1-\alpha)(\theta^h-\theta^l)}{(e+1)^2}\left(\eta-\frac{2}{e+1}\right),$$

which is strictly negative for $e<\frac{2-\eta}{\eta}$ and strictly positive for $e>\frac{2-\eta}{\eta}$. Hence, $\beta_h{}'(e)+\eta\beta_h(e)$ is strictly quasiconvex. □

3. Contract design

3.1. Constraints in contract design

The constraints in the principal's profit maximization problem are the same as in the paper. With monetary disutility of effort these constraints read

$$\alpha(\beta_h(e)\underline{u}_1(e)+(1-\beta_h(e))\bar{u}_1(e))+(1-\alpha)(\beta_l(e)\underline{u}_2(e)+(1-\beta_l(e))\bar{u}_2(e))$$
$$\geq\beta\underline{u}_2(0)+(1-\beta)\bar{u}_2(0), \quad \text{(IGl)}$$

[2]The possible second solution of the first order condition has to be a local minimum because between any two local maxima, there has to be a local minimum. Also note that $\lim_{e\to\infty}U(e)=-\infty$ which implies that U has a maximum.

$$\alpha(\beta_h(e)\underline{u}_1(e) + (1-\beta_h(e))\bar{u}_1(e)) + (1-\alpha)(\beta_l(e)\underline{u}_2(e) + (1-\beta_l(e))\bar{u}_2(e)) \geq \beta\underline{u}_1(0) + (1-\beta)\bar{u}_1(0), \quad \text{(IGh)}$$

$$\begin{aligned}&\alpha\left(\beta_h(e)\underline{u}_1(e) + (1-\beta_h(e))\bar{u}_1(e)\right) + (1-\alpha)\left(\beta_l(e)\underline{u}_2(e) + (1-\beta_l(e))\bar{u}_2(e)\right)\\ \geq\ &\alpha\left(\beta_h(e^h)\underline{u}_1(e^h) + (1-\beta_h(e^h))\bar{u}_1(e^h)\right) + (1-\alpha)\left(\beta_l(e^h)\underline{u}_0(e^h) + (1-\beta_l(e^h))\bar{u}_0(e^h)\right),\end{aligned} \quad \text{(EH)}$$

$$\beta_l(e)\underline{u}_2(e) + (1-\beta_l(e))\bar{u}_2(e) \geq \beta_l(e)\underline{u}_0(e) + (1-\beta_l(e))\bar{u}_0(e), \quad \text{(IRl)}$$

$$\beta_h(e)\underline{u}_1(e) + (1-\beta_h(e))\bar{u}_1(e) \geq \beta_h(e)\underline{u}_2(e) + (1-\beta_h(e))\bar{u}_2(e), \quad \text{(ICh)}$$

where e denotes the optimal effort given that the agent buys contract 1 in case of signal σ^h and contract 2 in case of signal σ^l. Effort e^h denotes the optimal effort when one buys contract 1 in case of signal σ^h and no insurance in case of signal σ^l.

In the paper, we could neglect the following deviation of the agent: Exert effort e^l and buy contract 2 if the signal is σ^h and do not buy insurance if the signal is σ^l. To illustrate, let us assume $e^l < e$. Then, this deviation was ruled out by (IRl). Because $\beta^l(e) < \beta^l(e^l)$ an agent with a bad signal would find it optimal to buy contract 2 instead of remaining uninsured after exerting effort e^l. In the monetary cost framework, this is no longer straightforward. The problem is that the utilities depend on e. If the utility function exhibits strongly decreasing absolute risk aversion and e^l is much lower than e, a σ^l agent might find it better to stay uninsured after exerting effort e^l. The reason is that exerting an effort much below e makes him richer and therefore less risk averse. For this reason, we have to introduce the additional constraint

$$\begin{aligned}&\alpha\left(\beta_h(e)\underline{u}_1(e) + (1-\beta_h(e))\bar{u}_1(e)\right) + (1-\alpha)\left(\beta_l(e)\underline{u}_2(e) + (1-\beta_l(e))\bar{u}_2(e)\right)\\ \geq\ &\alpha\left(\beta_h(e^l)\underline{u}_2(e^l) + (1-\beta_h(e^l))\bar{u}_2(e^l)\right) + (1-\alpha)\left(\beta_l(e^l)\underline{u}_0(e^l) + (1-\beta_l(e^l))\bar{u}_0(e^l)\right)\end{aligned} \quad \text{(EL)}$$

when we want to deal with the general framework. However, it is straightforward that condition S1 allows us to use the same reasoning as in the paper to rule out that (EL) binds. Wealth effects can be neglected with CARA preferences.

For similar reasons, we also have to track the ex ante individual rationality constraint

$$\alpha(\beta_h(e)\underline{u}_1(e) + (1-\beta_h(e))\bar{u}_1(e)) + (1-\alpha)(\beta_l(e)\underline{u}_2(e) + (1-\beta_l(e))\bar{u}_2(e)) \geq \beta\underline{u}_0(0) + (1-\beta)\bar{u}_0(0). \quad \text{(EAIR)}$$

In the setting of the paper, this constraint was implied by (IRl) and (IGl): Given that a low signal agent preferred contract 2 to the outside option, also an agent exerting zero

effort (and therefore having a risk higher than the low signal agent) preferred contract 2 to the outside option. Again this is no longer clear in the monetary effort cost setup: Risk aversion after exerting effort $e > 0$ could be different from risk aversion when not exerting effort. Therefore, the preferences over contract 2 and contract 0 could depend on the level of exerted effort. With CARA preferences the reasoning from the paper goes through as risk aversion does not depend on exerted effort.

As in the paper, some constraints are implied by others. First we present an intermediate result for which we do not have to use condition S1.

Lemma S2. *(ICh) is implied by (IGl). (IRl) is generically implied by (EH) (and it is always implied under condition 1).*

Proof. First consider the claim that (ICh) is implied by (IGl). Using $\beta = \alpha\beta_h(e) + (1-\alpha)\beta_l(e)$ and $1-\beta = \alpha(1-\beta_h(e)) + (1-\alpha)(1-\beta_l(e))$, (IGl) can be written as

$$\begin{aligned}\alpha\beta^h(e)\left(\underline{u}_1(e) - \underline{u}_2(0)\right) + \alpha(1-\beta^h(e))\left(\bar{u}_1(e) - \bar{u}_2(0)\right)\\ + (1-\alpha)\beta^l(e)\left(\underline{u}_2(e) - \underline{u}_2(0)\right) + (1-\alpha)(1-\beta^l(e))\left(\bar{u}_2(e) - \bar{u}_2(0)\right) \geq 0.\end{aligned}$$

Since the terms in the second line are negative ($\underline{u}_2(e) < \underline{u}_2(0)$ and $\bar{u}_2(e) < \bar{u}_2(0)$), the terms in the first line must be positive:

$$\alpha\beta_h(e)\left(\underline{u}_1(e) - \underline{u}_2(0)\right) + \alpha(1-\beta_h(e))\left(\bar{u}_1(e) - \bar{u}_2(0)\right) > 0.$$

Canceling α and using $\underline{u}_2(e) < \underline{u}_2(0)$ as well as $\bar{u}_2(e) < \bar{u}_2(0)$ gives

$$\beta_h(e)\left(\underline{u}_1(e) - \underline{u}_2(e)\right) + (1-\beta_h(e))\left(\bar{u}_1(e) - \bar{u}_2(e)\right) > 0,$$

which implies (ICh).

The second part of the proof is by contradiction. Suppose (IRl) was binding. Then (EH) could be written as $f(e) \geq f(e^h)$, where

$$f(e) = \alpha\left(\beta_h(e)\underline{u}_1(e) + (1-\beta_h(e))\bar{u}_1(e)\right) + (1-\alpha)\left(\beta_l(e)\underline{u}_0(e) + (1-\beta_l(e))\bar{u}_0(e)\right).$$

Note that e^h is defined as the maximizer of f and therefore $f(e^h) \geq f(e)$. Hence, we would get that e and e^h are both global maximizers of f. Since e is derived from the effort choice problem with contracts 1 and 2, i.e. from (3) and (4), in which $\bar{u}_0$ and $\underline{u}_0$ do not play a role, e will generically not be a global maximizer of f (contradiction in a generic sense!). With the structure imposed by condition S1 an even stronger result is obtained: It follows from condition S1 that f can only have one interior maximizer which is e^h. Hence, we get a contradiction. □

Lemma S3. *Assume condition S1 holds. Then (EAIR) and (EL) is implied by other constraints and (IGh) is slack in the optimal contract menu. (IGl) and (EH) are binding in the optimal contract menu.*

Proof. First consider the claim that (EAIR) is implied by other constraints. (IRl) and $\beta > \beta_l(e)$ imply

$$\beta(\underline{u}_2(e) - \underline{u}_0(e)) + (1-\beta)(\bar{u}_2(e) - \bar{u}_0(e)) \geq 0.$$

Using CARA preferences and multiplying through by $\mathbf{e}^{-\eta e}$, gives

$$\beta(-\mathbf{e}^{-\eta(w-p_2-D+R_2)} + \mathbf{e}^{-\eta(w-D)}) + (1-\beta)(-\mathbf{e}^{-\eta(w-p_2)} + \mathbf{e}^{-\eta w}) \geq 0.$$

This is equivalent to

$$\beta\underline{u}_2(0) + (1-\beta)\bar{u}_2(0) \geq \beta\underline{u}_0(0) + (1-\beta)\bar{u}_0(0).$$

Together with (IGl), the last equation implies (EAIR).

To show that (EL) is slack, we distinguish two cases. First, $e^l \leq e$. Then multiplying (IRl) through with $\mathbf{e}^{-\eta e}$ yields

$$\beta_l(e)(-\mathbf{e}^{-\eta(w-p_2-D+R_2)} + \mathbf{e}^{-\eta(w-D)}) + (1-\beta_l(e))(-\mathbf{e}^{-\eta(w-p_2)} + \mathbf{e}^{-\eta w}) \geq 0.$$

Using $\beta_l(e) \leq \beta_l(e^l)$ and multiplying through with $\mathbf{e}^{\eta e^l}$ gives

$$\beta_l(e^l)(-\mathbf{e}^{-\eta(w-p_2-D+R_2-e^l)} + \mathbf{e}^{-\eta(w-D-e^l)}) + (1-\beta_l(e^l))(-\mathbf{e}^{-\eta(w-p_2-e^l)} + \mathbf{e}^{-\eta(w-e^l)}) \geq 0.$$

Therefore,

$$\beta_l(e^l)\underline{u}_2(e^l) + (1-\beta_l(e^l))\bar{u}_2(e^l) \geq \beta_l(e^l)\underline{u}_0(e^l) + (1-\beta_l(e^l))\bar{u}_0(e^l).$$

Hence, the right-hand side of (EL) satisfies the following:

$$\begin{aligned}
& \alpha\left(\beta_h(e^l)\underline{u}_2(e^l) + (1-\beta_h(e^l))\bar{u}_2(e^l)\right) + (1-\alpha)\left(\beta_l(e^l)\underline{u}_0(e^l) + (1-\beta_l(e^l))\bar{u}_0(e^l)\right) \\
\leq\ & \alpha\left(\beta_h(e^l)\underline{u}_2(e^l) + (1-\beta_h(e^l))\bar{u}_2(e^l)\right) + (1-\alpha)\left(\beta_l(e^l)\underline{u}_2(e^l) + (1-\beta_l(e^l))\bar{u}_2(e^l)\right) \\
\leq\ & \beta\underline{u}_2(0) + (1-\beta)\bar{u}_2(0),
\end{aligned}$$

where the last inequality is strict for $e^l > 0$. Therefore (EL) is implied by (IGl).

Second, $e^l > e$. In this case, (ICh) multiplied through by $\mathbf{e}^{-\eta e}$ yields

$$\beta_h(e)(-\mathbf{e}^{-\eta(w-p_1-D+R_1)} + \mathbf{e}^{-\eta(w-p_2-D+R_2)}) + (1-\beta_h(e))(-\mathbf{e}^{-\eta(w-p_1)} + \mathbf{e}^{-\eta(w-p_2)}) \geq 0.$$

Using $\beta_h(e^l) > \beta_h(e)$ and multiplying through with $\mathbf{e}^{\eta e^l}$, gives

$$\beta_h(e^l)(-\mathbf{e}^{-\eta(w-p_1-D+R_1-e^l)} + \mathbf{e}^{-\eta(w-p_2-D+R_2-e^l)}) + (1-\beta_h(e^l))(-\mathbf{e}^{-\eta(w-p_1-e^l)} + \mathbf{e}^{-\eta(w-p_2-e^l)}) \geq 0.$$

Therefore, the right-hand side of (EL) satisfies the following:

$$\begin{aligned}
& \alpha\left(\beta_h(e^l)\underline{u}_2(e^l)+(1-\beta_h(e^l))\bar{u}_2(e^l)\right)+(1-\alpha)\left(\beta_l(e^l)\underline{u}_0(e^l)+(1-\beta_l(e^l))\bar{u}_0(e^l)\right) \\
\leq\ & \alpha\left(\beta_h(e^l)\underline{u}_1(e^l)+(1-\beta_h(e^l))\bar{u}_1(e^l)\right)+(1-\alpha)\left(\beta_l(e^l)\underline{u}_0(e^l)+(1-\beta_l(e^l))\bar{u}_0(e^l)\right) \\
\leq\ & \alpha\left(\beta_h(e^h)\underline{u}_1(e^h)+(1-\beta_h(e^h))\bar{u}_1(e^h)\right)+(1-\alpha)\left(\beta_l(e^h)\underline{u}_0(e^h)+(1-\beta_l(e^h))\bar{u}_0(e^h)\right),
\end{aligned}$$

where the last inequality follows from the definition of e^h as the effort level maximizing expected utility when deciding between contracts 1 and 0. Consequently, (EH) implies (EL).

Now we turn to the claim about (IGh). Suppose (IGh) was binding. We will show that the principal has a deviation pooling contract that satisfies all constraints and increases profits. Hence, the initial menu cannot be optimal. We distinguish two cases.

First, suppose $\bar{u}_1(e) > \underline{u}_1(e)$. In this case, we claim that profits can be increased by only offering contract 1, i.e. by dropping contract 2 from the menu. Since (IGh) was binding, the agent can achieve the same utility as before by buying contract 1 without exerting effort. This must be his optimal choice as restricting his choice set cannot result in higher ex ante utility. In particular, (EAIR) and (EH) are not affected and (IGl) is irrelevant in the pooling situation. Now we have to show that profits are increased:

$$\begin{aligned}
\pi^{deviation} &= w-\beta D-\beta h(\underline{u}_1(0))-(1-\beta)h(\bar{u}_1(0)) \\
&= w-\beta D-\alpha\left(\beta_h(e)h(\underline{u}_1(0))+(1-\beta_h(e))h(\bar{u}_1(0))\right) \\
&\quad -(1-\alpha)\left(\beta_l(e)h(\underline{u}_1(0))+(1-\beta_l(e))h(\bar{u}_1(0))\right) \\
&= w-\beta D-e-\alpha\left(\beta_h(e)h(\underline{u}_1(e))+(1-\beta_h(e))h(\bar{u}_1(e))\right) \\
&\quad -(1-\alpha)\left(\beta_l(e)h(\underline{u}_1(e))+(1-\beta_l(e))h(\bar{u}_1(e))\right) \\
&> w-\beta D-e-\alpha\left(\beta_h(e)h(\underline{u}_1(e))+(1-\beta_h(e))h(\bar{u}_1(e))\right) \\
&\quad -(1-\alpha)\left(\beta_l(e)h(\underline{u}_2(e))+(1-\beta_l(e))h(\bar{u}_2(e))\right) \\
&= \pi^{initial},
\end{aligned}$$

where the inequality follows from the following observation: We know that $\beta_l(e)\underline{u}_2(e)+(1-\beta_l(e))\bar{u}_2(e) \geq \beta_l(e)\underline{u}_1(e)+(1-\beta_l(e))\bar{u}_1(e)$. Let us assume for now that this inequality held with equality. Because h is strictly convex and $\underline{u}_2(e) < \underline{u}_1(e) < \bar{u}_1(e) < \bar{u}_2(e)$, the line connecting the points $(\underline{u}_1(e), h(\underline{u}_1(e)))$ and $(\bar{u}_1(e), h(\bar{u}_1(e)))$ is strictly below the line connecting the points $(\underline{u}_2(e), h(\underline{u}_2(e)))$ and $(\bar{u}_2(e), h(\bar{u}_2(e)))$. If—as we assumed for now—$\beta_l(e)\underline{u}_2(e)+(1-\beta_l(e))\bar{u}_2(e) = \beta_l(e)\underline{u}_1(e)+(1-\beta_l(e))\bar{u}_1(e)$, this implies that $\beta_l(e)h(\underline{u}_2(e))+(1-\beta_l(e))h(\bar{u}_2(e)) > \beta_l(e)h(\underline{u}_1(e))+(1-\beta_l(e))h(\bar{u}_1(e))$. If the inequality

$\beta_l(e)\underline{u}_2(e)+(1-\beta_l(e))\bar{u}_2(e) \geq \beta_l(e)\underline{u}_1(e)+(1-\beta_l(e))\bar{u}_1(e)$ holds strictly, the inequality $\beta_l(e)h(\underline{u}_2(e))+(1-\beta_l(e))h(\bar{u}_2(e)) > \beta_l(e)h(\underline{u}_1(e))+(1-\beta_l(e))h(\bar{u}_1(e))$ is satisfied even clearer.[3]

Second, suppose $\bar{u}_1(e) \leq \underline{u}_1(e)$. Then a full coverage pooling contract that gives the same ex ante utility to the agent increases profits and is feasible. Let $u_p(0) = \alpha(\beta_h(e)\underline{u}_1(e)+(1-\beta_h(e))\bar{u}_1(e))+(1-\alpha)(\beta_l(e)\underline{u}_2(e)+(1-\beta_l(e))\bar{u}_2(e))$ be the utility level of the full coverage deviation contract. The agent can achieve the same utility level as before by buying the pooling contract without exerting effort. Hence, (EAIR) is satisfied (and (IGl) is irrelevant in pooling). Because (IGh) was binding initially by assumption, we have $u_p(0) = \beta\underline{u}_1(0)+(1-\beta)\bar{u}_1(0)$. As $\beta_h(e') > \beta$ for any $e' > 0$ and $\bar{u}_1(0) \leq \underline{u}_1(0)$ by the assumption $\bar{u}_1(e) \leq \underline{u}_1(e)$ and CARA, it follows that $u_p(0) \leq \beta_h(e')\underline{u}_1(0)+(1-\beta_h(e'))\bar{u}_1(0)$. Using the CARA preferences and multiplying this inequality through with $\mathbf{e}^{\eta e'}$, leads to the inequality $u_p(e') \leq \beta_h(e')\underline{u}_1(e')+(1-\beta_h(e'))\bar{u}_1(e')$. Therefore, the right hand side of (EH) is (weakly) lower in the deviation menu than in the initial menu for any effort level. Since the left hand side of (EH) is unchanged by the definition of u_p, (EH) is (weakly) relaxed by the deviation. Finally, we show that profits are higher under the deviation contract:

$$
\begin{aligned}
\pi^{deviation} &= w-\beta D-h(u_p(0)) \\
&= w-\beta D \\
&\quad -h\left(\alpha(\beta_h(e)\underline{u}_1(e)+(1-\beta_h(e))\bar{u}_1(e))+(1-\alpha)(\beta_l(e)\underline{u}_2(e)+(1-\beta_l(e))\bar{u}_2(e))\right) \\
&> w-\beta D-\alpha\left(\beta_h(e)h(\underline{u}_1(e))+(1-\beta_h(e))h(\bar{u}_1(e))\right) \\
&\quad -(1-\alpha)\left(\beta_l(e)h(\underline{u}_2(e))+(1-\beta_l(e))\bar{u}_2(e)\right) \\
&\geq w-\beta D-e-\alpha\left(\beta_h(e)h(\underline{u}_1(e))+(1-\beta_h(e))h(\bar{u}_1(e))\right) \\
&\quad -(1-\alpha)\left(\beta_l(e)h(\underline{u}_2(e))+(1-\beta_l(e))\bar{u}_2(e)\right) \\
&= \pi^{initial}
\end{aligned}
$$

where the first inequality follows from the strict convexity of h and the second inequality follows from $e \geq 0$.

Last we turn to the binding constraints. Until now we proved that only (IGl) and (EH) can be binding. Suppose (IGl) was not binding. Then marginally decreasing R_1 increases profits and relaxes the only binding constraint (EH).[4] Similarly, suppose (EH) was slack. Then increasing both p_1 and p_2 by $\varepsilon > 0$ will increase profits without

[3]Note that we did not use condition S1 for this part of the proof. Hence, the statement "(IGh) cannot bind if $\bar{u}_1(e) > \underline{u}_1(e)$" holds in general.

[4]This assumes $e^h > e$ as one should expect. As we did not prove this: If $e^h \leq e$, increasing R_1 and p_1 by $\varepsilon > 0$ increases profits and relaxes (EH).

affecting the only binding constraint (IGl) (this follows from the CARA preferences). It follows that both (IGl) and (EH) must be binding. □

Before proceeding to optimal contracts, we still need to formally show that $e^h \geq e$, which is intuitively obvious but no longer as straightforward as in the paper as the optimal effort level depends not only on the Δ_is but also on utility levels.

Lemma S4. *Assume condition S1 holds. Then $e^h \geq e$.*

Proof. The proof is by contradiction. Suppose $e^h < e$. By (IRl) ,

$$\beta_l(e)(-\mathbf{e}^{-\eta(w-p_2-D+R_2-e)} + \mathbf{e}^{-\eta(w-D-e)}) + (1-\beta_l(e))(-\mathbf{e}^{-\eta(w-p_2-e)} + \mathbf{e}^{-\eta(w-e)}) \geq 0.$$

Multiplying through with $\mathbf{e}^{-\eta(e-e^h)}$ and using $\beta_l(e^h) > \beta_l(e)$ gives

$$\beta_l(e^h)(-\mathbf{e}^{-\eta(w-p_2-D+R_2-e^h)} + \mathbf{e}^{-\eta(w-D-e^h)}) + (1-\beta_l(e^h))(-\mathbf{e}^{-\eta(w-p_2-e^h)} + \mathbf{e}^{-\eta(w-e^h)}) > 0,$$

which is equivalent to

$$\beta_l(e^h)\underline{u}_2(e^h) + (1-\beta_l(e^h))\bar{u}_2(e^h) > \beta_l(e^h)\underline{u}_0(e^h) + (1-\beta_l(e^h))\bar{u}_0(e^h). \tag{6}$$

By Lemma S3, (EH) is binding. Therefore,

$$\begin{aligned}
&\alpha\left(\beta_h(e)\underline{u}_1(e) + (1-\beta_h(e))\bar{u}_1(e)\right) + (1-\alpha)\left(\beta_l(e)\underline{u}_2(e) + (1-\beta_l(e))\bar{u}_2(e)\right) \\
&= \alpha\left(\beta_h(e^h)\underline{u}_1(e^h) + (1-\beta_h(e^h))\bar{u}_1(e^h)\right) + (1-\alpha)\left(\beta_l(e^h)\underline{u}_0(e^h) + (1-\beta_l(e^h))\bar{u}_0(e^h)\right).
\end{aligned}$$

If we define g as

$$g(e) = \alpha\left(\beta_h(e)\underline{u}_1(e) + (1-\beta_h(e))\bar{u}_1(e)\right) + (1-\alpha)\left(\beta_l(e)\underline{u}_2(e) + (1-\beta_l(e))\bar{u}_2(e)\right),$$

then the last equation yields $g(e) < g(e^h)$ because of (6). This contradicts the definition of e as the effort level maximizing the left hand side of this inequality. □

3.2. Optimal contract

The main result we want to show here is that both contracts have less than full coverage.

Proposition S1. *Assume condition S1. The optimal contract menu includes only partial coverage contracts, i.e. $\bar{u}_1(e) - \underline{u}_1(e) > 0$.*

Proof. We first prove the proposition for a separating equilibrium. Suppose $R_1 \geq D$, i.e. (more than) full coverage. We will show that the principal has a profitable deviation in this case. Using the implicit function theorem on (5), we get

$$\begin{aligned}
\frac{d\,e}{dp_1} &= \frac{-\eta^2\alpha\mathbf{e}^{\eta p_1} + \alpha\eta(\beta_h{}'(e) + \eta\beta_h(e))(\mathbf{e}^{\eta p_1} - \mathbf{e}^{\eta(p_1+D-R_1)})}{-SOC}, \\
\frac{d\,e}{dR_1} &= \frac{\eta\alpha\mathbf{e}^{\eta(p_1+D-R_1)}\left(\eta\beta_h(e) + \beta_h{}'(e)\right)}{-SOC} > 0,
\end{aligned}$$

where SOC is the derivative of (5) with respect to e, which must be negative by the second order condition. Now consider changing contract 1 marginally while keeping contract 2 fixed. The principal's profits are $\alpha p_1 + (1-\alpha)p_2 - \alpha\beta_h(e)R_1 - (1-\alpha)\beta_l(e)R_2$. The slope of the principal's isoprofit curve in the (R_1, p_1) plane is

$$\left.\frac{d\,p_1}{dR_1}\right|_{\pi=const} = \frac{\beta_h(e) + \frac{d\,e}{dR_1}\beta_h{}'(e)\,(R_1 - R_2)}{1 - \frac{d\,e}{dp_1}\beta_h{}'(e)\,(R_1 - R_2)} > \beta_h(e),$$

where the last inequality follows from the following steps: $1 - \beta_h(e) - (\beta_h'(e))/\eta < 0$ because rearranging (5) gives[5]

$$\begin{aligned}
\beta_h(e) + \frac{\beta_h'(e)}{\eta} &= \frac{\alpha\left(\mathbf{e}^{-\eta(-p_1)} - \mathbf{e}^{-\eta(-p_2)}\right) + \beta\mathbf{e}^{-\eta(-p_2-D+R_2)} + (1-\beta)\mathbf{e}^{-\eta(-p_2)}}{-\alpha\mathbf{e}^{-\eta(-p_2)} + \alpha\mathbf{e}^{-\eta(-p_2-D+R_2)} + \alpha\mathbf{e}^{-\eta(-p_1)} - \alpha\mathbf{e}^{-\eta(-p_1-D+R_1)}} \\
&\geq \frac{\alpha\left(\mathbf{e}^{-\eta(-p_1)} - \mathbf{e}^{-\eta(-p_2)}\right) + \mathbf{e}^{-\eta(-p_2-D+R_2)}}{\alpha\left(\mathbf{e}^{-\eta(-p_1)} - \mathbf{e}^{-\eta(-p_2)}\right) + \alpha\mathbf{e}^{-\eta(-p_2-D+R_2)} - \alpha\mathbf{e}^{-\eta(-p_1-D+R_1)}} \\
&> \frac{\alpha\left(\mathbf{e}^{-\eta(-p_1)} - \mathbf{e}^{-\eta(-p_2)}\right) + \mathbf{e}^{-\eta(-p_2-D+R_2)}}{\alpha\left(\mathbf{e}^{-\eta(-p_1)} - \mathbf{e}^{-\eta(-p_2)}\right) + \mathbf{e}^{-\eta(-p_2-D+R_2)}} = 1.
\end{aligned}$$

But then using the expressions above, we have

$$\begin{aligned}
0 &> \mathbf{e}^{\eta p_1}(R_1 - R_2)\,(1 - \beta_h(e) - (\beta_h'(e))/\eta) \\
\Leftrightarrow \frac{d\,e}{dR_1}\beta_h{}'(e)\,(R_1 - R_2) &> -\frac{d\,e}{dp_1}\beta_h{}'(e)\,(R_1 - R_2)\,.
\end{aligned}$$

This gives $\left.\frac{d\,p_1}{dR_1}\right|_{\pi=const} > \beta_h(e)$

Similarly, we derive the slope of the ex ante indifference curve of the agent. Using an envelope argument and canceling terms leads to

$$\left.\frac{d\,p_1}{dR_1}\right|_{U=const} = \frac{\beta_h(e)}{\beta_h(e) + (1-\beta_h(e))\mathbf{e}^{\eta(R_1-D)}} \leq \beta_h(e),$$

where the inequality follows from $R_1 \geq D$.

This shows that reducing R_1 by $\varepsilon > 0$ and p_1 by $\varepsilon \left.\frac{d\,p_1}{dR_1}\right|_{U=const}$ will increase the principal's payoff while leaving the agent's payoff unaffected (for $\varepsilon > 0$ small enough). The binding constraint (IGl) is not affected by this change while the constraint (EH) is even relaxed: The slope of the indifference curve (in (R_1, p_1) space) of an agent exerting effort e^h is $\frac{\beta_h(e^h)}{\beta_h(e^h)+(1-\beta_h(e^h))\mathbf{e}^{\eta(R_1-D)}}$ which is steeper than the indifference curve

[5]The step from the first to the second line assumes that $D \geq R_2$, i.e., that contract 2 provides at most full coverage. It is easy to show that this must be true: If both contracts had more than full coverage, the insurer could make higher profits by offering a full coverage pooling contract that gives the same expected utility to the agent as in the original situation. As coverage is lower in this deviation contract, (EH) is relaxed. As the agent's expected utility is the same, (IGl) is unaffected. As the contract is more efficient (full coverage, no effort), it increases the insurer's profits.

of an agent exerting effort e as $\beta_h(e^h) \geq \beta_h(e)$. Hence, the deviation is feasible and profitable.

The proof for a pooling equilibrium is similar. In a pooling equilibrium, $e = 0$ and $\frac{de}{dp} = \frac{de}{dR} = 0$. For $R \geq D$, we consequently get

$$\left.\frac{dp}{dR}\right|_{\pi=const} = \beta \geq \frac{\beta}{\beta + (1-\beta)\mathbf{e}^{\eta(R-D)}} = \left.\frac{dp}{dR}\right|_{U=const}.$$

Furthermore, the slope of the indifference curve of an agent exerting effort $e^h > 0$ (and buying insurance only when receiving a high signal) is $\frac{\beta_h(e^h)}{\beta_h(e^h)+(1-\beta_h(e^h))\mathbf{e}^{\eta(R-D)}}$, which is strictly higher than $\frac{\beta}{\beta+(1-\beta)\mathbf{e}^{\eta(R-D)}}$ as $\beta_h(e^h) > \beta$. Consequently, reducing R_1 by $\varepsilon > 0$ and p_1 by $\frac{\varepsilon\beta}{\beta+(1-\beta)\mathbf{e}^{\eta(R-D)}}$ will (i) keep the utility of the agent constant, (ii) (weakly) increase the principal's profit and (iii) strictly relax the only binding constraint (EH). As the only binding constraint is strictly relaxed, decreasing R_1 by slightly more than ε is still feasible and strictly increases the principal's payoff. □

Given condition S1, we know that (IGl) and (EH) are binding. This implies immediately Proposition 4 in the paper.[6] Also the two binding constraints are again linear in utilities. The first order conditions for e and e^h (as in (3)) are linear in utilities for CARA preferences. Therefore, it is straightforward to solve these four conditions for the utility levels $\bar{u}_1(e)$, $\underline{u}_1(e)$, $\bar{u}_2(e)$ and $\underline{u}_2(e)$. Using these expressions we can write the principal's maximization as a maximization problem over e and e^h under the constraint $e^h \geq e \geq 0$ which implies an equivalent of Proposition 2 in the paper. If we use γe instead of e in the monetary effort cost setup and let γ approach 0, the Stiglitz menu results (see Proposition 3 in the paper). This is true as the proof in the paper only relies on continuity of the utility/cost function which is unproblematic in the monetary effort cost framework.

[6] In the monetary effort cost setup, we would use $-\gamma e$ instead of $-e$ in the argument of the utility function to parameterize costs.